Xiaole "Alex" Jiang

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Summary

Robot learning researcher working on **world models and planning for visuomotor control**. Came to robotics from theoretical physics (quantum gravity Ph.D., Jan 2027). Since then, proposed and led k-WM, which lets world models predict many future steps at once for faster, more reliable long-horizon planning (ICLR 2027 submission), and built the visuo-tactile manipulation policies for PanoTac, a new dense tactile sensor (ICRA 2027 submission); co-first author on both. Triple major in physics, mathematics, and philosophy; always happy to talk ideas across fields.

Research Experience

NYU Generalizable Robotics and AI Lab (GRAIL), Courant Institute May 2025 – Present
Visiting Researcher *New York, NY*

k-WM: Predicting k Steps at Once for Long-Horizon Planning with World Models

X. A. Jiang*, Y. Wang*, L. Pinto, Y. LeCun, G. Zhou. *Under review, ICLR 2027.*

- Proposed and led the project: a training objective that lets transformer-based JEPA world models predict a variable-length sequence of future latent states in a single forward pass.
- Raised long-horizon planning success by 37% (gradient-based) and 15% (sampling-based) relative to baseline world models and cut planning time up to **3.3**×, across three world-model families and four control benchmarks.
- Showed that the multi-step objective induces a latent geometry mirroring the ground-truth environment.

PanoTac: A Dense Three-Axis Tactile Sensor for Robust Touch-Centric Manipulation

B. Wang*, **X. Jiang***, Z. Teng, X. Liu, Y. Li, L. Pinto, Z. J. Cui, B. Huang. *Under review, ICRA 2027.*

- Built the visuo-tactile diffusion-policy training and real-time inference pipeline on a Franka Panda, fusing wrist camera, dense touch, and proprioception: **86.7%** success vs. 43.3% for vision-only across 60 real-robot rollouts.

YOR: Your Own Mobile Manipulator for Generalizable Robotics

M. H. Anjaria et al., incl. **X. Jiang**, S. Chintala, L. Pinto. *arXiv:2602.11150, 2026.*

- Built the whole-body teleoperation, data collection, data processing, and policy training stack for a sub-$10K open-source mobile bimanual robot; validated the platform by training a behavior-cloning policy.

The Graduate Center, City University of New York 2021 – Present
Ph.D. Researcher, Theoretical Physics (advisor: Prof. Daniel Kabat) *New York, NY*

- Quantum gravity and holography: computed perturbative corrections to modular Hamiltonians in field theory.
- Quantum information: developed a group-theoretic framework for classifying multipartite entanglement.

Education

The Graduate Center, City University of New York Expected Jan 2027
Ph.D. in Theoretical Physics: quantum gravity, holography (AdS/CFT), entanglement *CUNY Science Fellowship*

Bates College 2017 – 2021
Triple major in Physics, Mathematics, and Philosophy; magna cum laude *Physics & Astronomy Prize; Philosophy Prize*
Honors thesis: Convection in Rotating Spherical Fluid Shell (Navier–Stokes simulations on a compute cluster)

Publications

* equal contribution. High-energy theory papers list authors alphabetically.

Robot Learning & Machine Learning

[1] **X. A. Jiang***, Y. Wang*, L. Pinto, Y. LeCun, G. Zhou. **k-WM: Predicting k Steps at Once for Long-Horizon Planning with World Models.** *Under review, ICLR 2027.* [website]

[2] B. Wang*, **X. Jiang***, Z. Teng, X. Liu, Y. Li, L. Pinto, Z. J. Cui, B. Huang. **PanoTac: A Dense Three-Axis Tactile Sensor for Robust Touch-Centric Manipulation.** *Under review, ICRA 2027.* [website]

[3] M. H. Anjaria, M. E. Erciyes, V. Ghatnekar, N. Navarkar, H. Etukuru, **X. Jiang**, K. Patel, D. Kabra, N. Wojno, R. A. Prayage, S. Chintala, L. Pinto, N. M. M. Shafiullah, Z. J. Cui. **YOR: Your Own Mobile Manipulator for Generalizable Robotics.** *arXiv:2602.11150, 2026.* [arXiv]

Physics

[4] **X. Jiang**, D. Kabat, A. Marthandan, D. Sarkar. **Modular Hamiltonians for Future-Perturbed States.** *Journal of High Energy Physics* 02 (2026) 210. [doi]

[5] **X. Jiang**, D. Kabat, G. Lifschytz, A. Marthandan. **Entanglement Groups.** *Annals of Physics* 494 (2026) 170634. [doi]

[6] M. Abdullah, C. Bavor, B. Chafamo, **X. Jiang**, M. H. Kalim, K. Stiffler, C. A. Whiting. **Inflation from Dynamical Projective Connections.** *Physical Review D* 106 (2022) 084049. [doi]

[7] N. Lundblad, R. A. Carollo, C. Lannert, M. J. Gold, **X. Jiang**, D. Paseltiner, N. Sergay, D. C. Aveline. **Shell Potentials for Microgravity Bose–Einstein Condensates.** *npj Microgravity* 5, 30 (2019). [doi]

Skills

Training and deploying visuomotor policies on real robots *(diffusion policy, behavior cloning; Franka, mobile bimanual)*
World models and model-based planning *(JEPA, MPC)*
Teleoperation and data-collection pipelines for robot learning
Visuo-tactile sensor fusion and real-time policy inference
Multi-GPU training and fine-tuning of transformer-based vision models *(ViTs)*
Software: Python, PyTorch, Git, Linux

Teaching, Awards & Service

- **Adjunct Lecturer in Physics**, Lehman College (CUNY): taught every semester, 2022 – present.
- **Best Visualization Award**, IAIFI Summer School Hackathon. Summer schools: Les Houches (Quantum Geometry, 2024); ICTP-SAIFR (Holography@25).
- **Board of Directors**, CUNY Research Foundation; Advancement & External Affairs Committee, CUNY Board of Trustees.